For your guide, explain the project in the same order in which it was developed: **simulation → vehicle model → path generation → control → perception → autonomous mission → disturbances → cooperative control → MACS GUI → safety/demo validation**.

You can use the following as your complete revision sheet.

1. Project objective

Main objective: Develop and validate an autonomous marine vehicle framework in simulation using **ROS 2 + Gazebo Harmonic**, capable of path following, vessel detection, autonomous approach/inspection, cooperative operation between two USVs, and operator supervision through MACS.

A good one-line explanation is:

"The project builds a ROS 2–Gazebo autonomous surface vehicle framework that integrates marine simulation, path planning, LOS guidance, PID/SMC control, AI-based vessel detection, autonomous mission logic, cooperative control, and a supervisory GUI."

---

2. Basic platform

Keyword	Meaning / what to explain
ROS 2	Robot Operating System 2. Middleware connecting sensors, controllers, perception, mission logic and GUI using nodes, topics, services and actions.
ROS 2 Humble	The ROS 2 distribution used in the project.
Gazebo Harmonic	3D simulator used for the virtual marine environment and USV models.
Gazebo Sim	Modern Gazebo simulation platform; this project is not Gazebo Classic.
WSL2	Windows Subsystem for Linux used to run Ubuntu and the ROS/Gazebo development environment.
Ubuntu 22.04	Linux distribution used for ROS 2 Humble.
Workspace	/home/sireesha/marine_ws, containing packages, scripts, launch files and tests.

Keyword	Meaning / what to explain
Node	Independent ROS program performing a specific function.
Topic	Continuous ROS communication channel, e.g. odometry, camera or control diagnostics.
Publisher	Node that sends data to a topic.
Subscriber	Node that receives topic data.
Service	Request/response interface, used for actions such as Safe Stop.
Action	Long-running command with feedback/result, used by MACS for mission execution.
TF	ROS coordinate-transform system connecting world, vehicle and sensor frames.

The architecture is approximately:

Gazebo

↓

Sensors / Vehicle State

↓

ROS 2 Topics

↓

Guidance

↓

Controller

↓

Thruster Commands

↓

Vehicle Motion

Camera → Perception → Mission Manager → Guidance/Controller

Cooperative Coordinator

↓

Leader + Follower

All major states

↓

MACS Operator GUI

---

3. Phase 1 — Marine environment

The first stage was simply proving that the marine simulation environment worked.

Keyword	Meaning
World	Gazebo environment containing water, lights and vehicles.
Calm Ocean	Base water environment used for physical simulation.
Water surface	Visual representation of the sea.
Buoyancy	Upward force keeping the vehicle floating.
Displacement	Amount of water displaced by the vehicle.
Hydrostatics	Forces determining how the vessel floats at rest.
Visual wave mesh	Wavy-looking water used for presentation only. It does **not** physically disturb the vehicle.
PBR	Physically Based Rendering, used to improve visual appearance of water/materials.
Normal map	Texture used to make the water appear to have fine wave detail.

Important distinction:

“The upgraded ocean surface is visual. Physical buoyancy is still based on the calm-water analytical model.”

---

4. Phase 2 — USV / catamaran model

Next, the autonomous surface vehicle was developed.

Keyword	Meaning
USV	Unmanned Surface Vehicle.
ASV	Autonomous Surface Vehicle.
Catamaran	Vessel with two parallel hulls.
Under-actuated vehicle	Vehicle with fewer independent actuators than motion degrees of freedom.
Surge	Forward/backward motion.
Sway	Sideways motion.
Heave	Vertical motion.
Roll	Rotation around longitudinal axis.
Pitch	Rotation around lateral axis.
Yaw	Rotation around vertical axis.
Differential thrust	Steering by applying different thrust values to left and right motors.
Port	Left side of vessel.
Starboard	Right side.
Thruster	Propulsion actuator generating force.
Direct-force thruster model	Simulation applies thrust force directly instead of modeling a detailed propeller.

Steering principle:

Port thrust = Starboard thrust

→ straight movement

Port > Starboard

→ turn one direction

Starboard > Port

→ turn opposite direction

Phase-2 performance was approximately:

Forward/reverse: ±0.79 m/s

Turning: ±0.39 rad/s

---

5. Vehicle localization and state

The controllers need to know where the vehicle is.

Keyword	Meaning
Odometry	Estimated position, orientation and velocity of the vehicle.
Pose	Position + orientation.
Twist	Linear and angular velocity.
IMU	Inertial Measurement Unit measuring orientation/angular motion/acceleration information.
Heading	Direction the vessel faces.
Yaw angle	Mathematical representation of heading.
Yaw rate	Rate at which heading changes.
Frame	Coordinate reference such as world, odom, or base_link.
base_link	Main coordinate frame attached to the vehicle.
odom	Local navigation reference frame.

Example:

/asv/odom

provides position and velocity to guidance/controller nodes.

---

6. Phase 3 — Path generation

After the vehicle worked, reference paths were generated.

Keyword	Meaning
Waypoint	Desired point along a route.
Reference path	Desired trajectory that the USV should follow.
Spline	Smooth mathematical curve through waypoints.
Cubic spline	Third-order polynomial interpolation producing smooth paths.
Natural cubic spline	Cubic spline with natural boundary conditions.
Straight path	Basic reference route.
Curved path	Smooth turning route.
S-shaped path	More challenging changing-curvature route.
Path progress	Percentage of the route completed.

Why splines?

“Directly connecting waypoints produces abrupt corners. Cubic splines generate smooth curvature that a marine vehicle can follow more realistically.”

---

7. LOS guidance

The path does not directly command the motors. A guidance law determines where the vessel should point.

Keyword	Meaning
LOS	Line-of-Sight guidance.
Look-ahead point	Point ahead on the path that the vehicle aims toward.
Desired heading	Heading generated by guidance.
Cross-Track Error / CTE	Perpendicular distance between vessel and reference path.
Heading error	Difference between desired and actual heading.

Conceptually:

Reference path

↓

Find closest / look-ahead point

↓

Generate desired heading

↓

Controller tries to achieve that heading

CTE is one of your main evaluation metrics.

---

8. Phase 4 — PID controller

The first closed-loop controller was PID.

Keyword	Meaning
PID	Proportional–Integral–Derivative controller.
Kp	Proportional gain; responds to present error.
Ki	Integral gain; responds to accumulated error.
Kd	Derivative gain; responds to error change.
Closed-loop control	Controller continuously compares desired and actual states and corrects the error.
Feedback	Measured state returned to the controller.

Final PID gains:

Kp = 0.5

Ki = 0.0

Kd = 0.6

Interpretation:

Kp → correct heading error

Kd → damp rapid heading changes

Ki = 0 → no accumulated-error contribution

Final approximate PID results:

Straight:

CTE RMSE ≈ 0.569 m

Curved:

CTE RMSE ≈ 0.574 m

S-shaped:

CTE RMSE ≈ 0.583 m

---

9. RMSE and evaluation metrics

Keyword	Meaning
RMSE	Root Mean Square Error. Measures overall tracking error magnitude.
CTE RMSE	Average effective cross-track tracking error across the run.
Heading RMSE	Overall heading-tracking error.
Tracking accuracy	How closely the vehicle follows the desired path.
Settling behavior	How the controller stabilizes after an error/disturbance.

Lower RMSE generally means better tracking.

---

10. SMC controller

A second controller was implemented for comparison.

Keyword	Meaning
SMC	Sliding Mode Control. Robust nonlinear control approach.
Sliding surface	Mathematical combination of tracking error and its derivative.
Switching control	Corrective control driving the state toward the sliding surface.
Boundary layer	Region used to reduce sharp switching/chattering.
Chattering	Rapid switching often associated with ideal SMC.

Keyword	Meaning
Robustness	Ability to maintain performance despite uncertainty/disturbances.

Selected SMC parameters:

lambda = 0.15

k = 0.05

phi = 0.08

omega = 0.5

Important answer if the guide asks which is better:

“We evaluated both PID and SMC. I do not claim SMC globally outperforms PID. The results depend on scenario and tuning; our SMC stability evidence is conditional/local.”

---

11. Controller mixer

A marine vehicle has left/right thrusters, while the controller deals with speed and turning.

Keyword	Meaning
Mixer	Converts forward/yaw demand into port and starboard thrust.
Surge command	Forward-speed demand.
Yaw command	Turning demand.
Saturation	Limiting thrust to allowed bounds.

Concept:

Forward command + turning command

↓

Mixer

↙ ↘

Port thrust Starboard thrust

---

12. Phase 5 — Camera and perception

The next stage added vessel perception.

Keyword	Meaning
Camera sensor	Simulated RGB/thermal-style image source.
Computer vision	Extracting useful information from images.
Object detection	Finding and classifying an object in an image.
Bounding box	Rectangle surrounding detected target.
Confidence	Detector certainty for a detection.
Inference	Running a trained neural-network model on an image.
FPS	Frames processed per second.
Image size / imgsz	Input resolution used by detector.

---

13. YOLO

Keyword	Meaning
YOLO	"You Only Look Once," a real-time object detection family.
YOLOv5	YOLO implementation used for the perception work.
YOLOv5n6	Nano P6 model used in the frozen detection pipeline.
P6	Architecture with an additional large-scale detection head.
Detection head	Part of model producing object predictions at a particular spatial scale.
Class	Object category predicted by network.
Weights	Learned neural-network parameters contained in .pt model.

Your model characteristics:

YOLOv5n6

10 classes

~3.11 million parameters

640 × 640 inference

4 detection heads

Important limitation:

“There was a class-mapping conflict between the dataset YAML and the trained checkpoint, so the final GUI deliberately reports a generic VESSEL target instead of claiming a specific ship subclass.”

---

14. RGB detection problem

The initial RGB trials produced no useful detections.

Important facts:

273 frames evaluated

0 detections

CPU throughput roughly 4.7–5 FPS

This was not hidden; it motivated another simulated sensing approach.

---

15. Thermal-style perception

The project then used an assigned-temperature simulated sensor.

Keyword	Meaning
Thermal-style image	Simulated image representing temperature differences.
Assigned temperature	Simulation assigns different thermal values to objects/background.
Pseudo-thermal	Thermal-like simulation rather than a calibrated real LWIR camera.
LWIR	Long-Wave Infrared. Real thermal-camera technology.

Important wording:

“It is a simulated assigned-temperature thermal-style system, not a physically calibrated LWIR thermal camera.”

---

16. Target bearing

Once a vessel is detected, its image position is converted into steering information.

Keyword	Meaning
Image center	Center pixel of camera image.
Target centroid	Center of detected bounding box.
Bearing error	Left/right angular error between camera center and target.
Visual servoing	Using image measurements to guide vehicle movement.

Concept:

Target left of image center

→ turn left

Target right

→ turn right

Target centered

→ approach forward

---

17. Target range

Your range is not produced by YOLO.

This distinction is very important.

Keyword	Meaning
Depth-derived range	Distance obtained independently from depth information.
Stand-off distance	Desired distance at which the vehicle stops from the target.
Arrival criterion	Rule declaring that the vessel is sufficiently close to the target.

Final mission stand-off:

12.0 m

Say:

“YOLO provides detection and image location. Range is independently depth-derived. YOLO itself does not measure distance.”

---

18. Search pattern

If the target is not initially visible, the vehicle performs a search.

Keyword	Meaning
Coverage search	Systematic movement to find an unknown target.
Three-lane search	Search trajectory consisting of three parallel coverage lanes.
Search path	Reference trajectory followed while looking for target.
Boundary deviation	Whether search trajectory leaves its allowed region.

Your validation included successful three-lane search missions.

---

19. Temporal confirmation

A single detector frame is not enough to immediately trigger the mission.

Keyword	Meaning
Temporal confirmation	Requiring detections over multiple frames/time before accepting target.
False positive	Incorrect detection.
Confirmation logic	Prevents reacting to one unstable prediction.

This makes the perception-to-mission transition more robust.

---

20. Mission state machine

The integrated mission uses an FSM.

Keyword	Meaning
FSM	Finite State Machine.
State	Current stage of autonomous mission.
Transition	Condition moving the system from one state to another.
Mission manager	Logic deciding what behavior should run next.

Keyword	Meaning
Arbitration	Deciding which command source currently has authority.

Typical flow:

SEARCH

↓

DETECTION

↓

CONFIRMATION

↓

ACQUIRE

↓

APPROACH

↓

ARRIVAL

↓

INSPECTION / MISSION_COMPLETE

Additional recovery states include:

TARGET_LOST

REACQUIRE

---

21. Target loss and reacquisition

Keyword	Meaning
Target loss	Detector temporarily no longer sees the vessel.
Reacquisition	Attempt to rediscover target.
Loss recovery	Logic preventing mission failure immediately after detection loss.
Mission hold	Intentionally stopping/holding propulsion while searching again.

During reacquisition, absence of a current range/confidence is not necessarily a fault.

---

22. Arrival and inspection

The vehicle approaches until it reaches the stand-off requirement.

Keyword	Meaning
Arrival	Vehicle satisfies target proximity criterion.
Zero propulsion	Thruster commands are commanded to zero.
Inspection state	Terminal mission condition after safe arrival.
Inspection image	Camera image recorded after propulsion stops.

Successful integrated mission reached approximately:

11.79 m

with a nominal target stand-off of:

12.0 m

---

23. Safety before inspection

A very useful point to explain:

Target detected

→ approach

→ stand-off reached

→ propulsion zero

→ inspection image/logging

The system does not simply continue driving into the target.

---

24. Disturbance validation

The framework was also evaluated under deterministic disturbances.

Keyword	Meaning
Disturbance	External influence that pushes the vehicle away from its intended behavior.

Keyword	Meaning
Deterministic disturbance	Same predefined disturbance can be repeated across tests.
Robustness test	Evaluating controller behavior under disturbances.
Calm condition	Baseline without disturbance.
Disturbed condition	Test with applied external effects.

Important limitation:

“The disturbance experiments are simulated deterministic forces. The visual ocean waves are not nonlinear physical waves.”

---

25. Cooperative control

Then the framework was extended from one USV to two.

Keyword	Meaning
Cooperative control	Multiple autonomous vehicles coordinating their motion.
Leader	Reference vehicle.
Follower	Vehicle trying to maintain a desired relationship with leader.
Formation	Desired geometric relationship between vehicles.
In-trail formation	Follower travels behind leader.
Formation spacing	Desired distance between vehicles.
Formation error	Difference between actual and desired spacing.

Your desired spacing:

8.0 m

Example:

Desired spacing = 8 m

Actual spacing = 5 m

Spacing error = -3 m

The follower then moves to correct that error.

---

26. Cooperative formation states

Keyword	Meaning
FORMATION_ACQUIRE	Follower is moving to obtain its desired formation position.
Formation tracking	Maintaining the required spacing once established.
Coordinator	Node managing the cooperative behavior.
Leader state	Information about leader pose/progress supplied to follower.
Communication degradation	Testing behavior when shared cooperative information becomes reduced/unavailable.

Your current framework supports **two vehicles**, not arbitrary large fleets.

---

27. Phase 8 vehicle-state relay

For each cooperative vessel:

/{vehicle}/odom

/{vehicle}/imu/raw

are converted/relayed into appropriate vehicle state/TF information.

Keyword	Meaning
Namespace	Separates leader and follower ROS topics.
Compatibility relay	Converts vehicle-specific state into the format expected by existing frozen controller.
Independent TF subtree	Leader and follower have separate coordinate-frame branches.

---

28. Feedback architecture

One of the strongest things to explain to your guide is that this is not an open-loop animation.

The information continuously circulates:

Vehicle motion

↓

Odometry / IMU

↓

Guidance

↓

Controller

↓

Thrusters

↓

Vehicle motion

That is a **closed-loop feedback system**.

For perception:

Camera

↓

YOLO

↓

Mission Manager

↓

Approach Command

↓

Vehicle

↓

New Camera Image

---

29. MACS

The final operator layer is MACS.

You can describe MACS as:

“MACS is the supervisory operator interface that combines navigation, perception, cooperative control, health monitoring and safety commands into a single GUI.”

Keyword	Meaning
MACS	Marine Autonomous Control/Supervisory operator interface used in the project.
GUI	Graphical User Interface.
Supervisory control	High-level monitoring/commands without directly taking over low-level thrusters.
AppState	Central GUI state representation.
RosBridge	Interface carrying ROS updates safely into Qt GUI thread.
Qt	GUI framework.
QThread	Separate thread used so ROS processing does not freeze the GUI.

---

30. MACS screens

MACS contains:

Overview

Navigation

Perception

Cooperative

Research

Settings

Overview

General vehicle status.

Navigation

Path, vessel position, CTE and guidance information.

Perception

Camera, detections, confidence and range.

Cooperative

Leader/follower spacing and formation information.

Research

Plots and evaluation data.

Settings

Operator/display configuration.

---

31. System health

Keyword	Meaning
Health aggregator	Combines subsystem states into an overall system-health assessment.
FRESH	Data is arriving within expected time.
STALE	Data has not updated recently enough.
LOST	Data has been absent beyond the expected threshold.
INVALID	Data exists but cannot currently be trusted/used.
NOT REQUIRED	Subsystem is intentionally unnecessary in current operating mode.
HEALTHY	Required systems operating correctly.
DEGRADED	Some noncritical capability impaired.
FAULT	Important required subsystem has failed.

A crucial point:

NOT_REQUIRED is not the same as a fault.

For PID path-following, for example, the camera may be NOT_REQUIRED.

---

32. Freshness engine

Keyword	Meaning
Freshness	How recently valid data was received.
Timestamp	Time associated with ROS data.

Keyword	Meaning
Timeout	Maximum acceptable time without an update.
Hysteresis / debounce	Prevents the GUI from rapidly switching state because of one short transient.

This was important because different mission phases naturally start and stop certain data streams.

---

33. Measurement availability versus stream freshness

This is another good technical concept.

The camera can be operating perfectly:

Camera stream = FRESH

while there is currently no vessel:

Confidence = NO CURRENT MEASUREMENT

Range = NO CURRENT MEASUREMENT

So MACS now distinguishes:

LIVE

NO CURRENT MEASUREMENT

LAST VALID

FINAL

INVALID

NOT REQUIRED

---

34. Last-valid/final data

If a range existed shortly before target loss:

12.1 m LAST VALID

After mission completion:

11.79 m FINAL

This is better than simply displaying —.

---

35. Operator-state model

MACS uses centralized state logic to avoid different parts of the GUI contradicting one another.

Important concepts:

mission_state

mission_active

mission_source

controller_state

can_start

can_reset

can_safe_stop

restart_supported

The GUI should not independently guess these values in every widget.

---

36. External mission detection

Sometimes run_marine_demo.sh starts the autonomous mission rather than MACS.

MACS therefore distinguishes:

MACS-managed mission

from:

Externally launched mission

If an external mission is already running:

START MISSION = LOCKED

This prevents a duplicate mission command.

---

37. Mission start gating

Before starting navigation, MACS checks prerequisites.

Examples:

ROS connected

Clock available

Odometry fresh

Controller ready

Safety clear

Valid path

No conflicting active mission

Guidance is **not required to already be publishing before mission start**.

After mission acceptance:

Guidance must appear within 2 seconds.

Otherwise:

GUIDANCE_START_TIMEOUT

This solved the circular start dependency.

---

38. Safe Stop

This is one of the most important safety features.

Keyword	Meaning
Safe Stop	Supervisory command that disables active propulsion and verifies zero thrust.
Latch	State indicating Safe Stop remains engaged.
Zero-propulsion verification	Confirms thruster commands have actually reached zero.
Coasting	Vessel may continue moving temporarily from momentum even with zero thrust.

Very important answer:

“Safe Stop guarantees zero propulsion, not instantaneous zero vessel velocity.”

The boat can still coast.

---

39. Safe Stop criteria

Your final design verifies approximately:

Port thrust ≈ 0

Starboard thrust ≈ 0

Controller inactive

Sustained for >= 0.5 s

with a small zero threshold around:

±0.01 N

---

40. Pause/Resume

The GUI does **not** pretend to support something the mission architecture cannot safely do.

Pause / Resume

→ intentionally unsupported

because the frozen mission system does not provide a true resumable hold interface.

This is a good design decision rather than a missing button.

---

41. Reset

Reset is only a supervisory reset.

It does not mean:

reset Gazebo

reset completed guidance

load a new path

restart every frozen node

Hence MACS describes it as:

RESET SUPERVISORY STATE ONLY

---

42. Completed mission restart limitation

A completed path cannot currently be replaced/reset through the frozen guidance interface in the same session.

Therefore after completion:

NEW SIMULATION SESSION REQUIRED

for another navigation demonstration.

This is why you use:

Fresh simulation

+

Fresh MACS

for every main demo.

---

43. ROS clock

Keyword	Meaning
/clock	Simulation time published by Gazebo/ROS bridge.
use_sim_time	Makes ROS nodes use simulation time instead of wall-clock time.
Sim time	Virtual time inside Gazebo.

If Gazebo resets while MACS remains running, timestamps can jump backward. That is why fresh MACS sessions are safer for demo transitions.

---

44. Parameter bridge

Keyword	Meaning
ros_gz_bridge	Connects Gazebo Transport messages and ROS 2 topics.
parameter_bridge	Bridge executable configured to translate selected Gazebo/ROS message types.

Example flow:

Gazebo sensor

→ Gazebo Transport

→ ros_gz_bridge

→ ROS 2 topic

---

45. RViz

Keyword	Meaning
RViz2	ROS visualization tool.
Path visualization	Shows planned/reference trajectory.
TF visualization	Shows coordinate frames.
Markers	Graphical ROS objects displayed in RViz.

Gazebo and RViz have different roles:

Gazebo → simulation/physics

RViz → ROS data visualization

---

46. Logging

Keyword	Meaning
Experiment logger	Records experiment data.
Inspection log	Stores final mission result/inspection information.
Evidence	Saved results proving that validation was performed.
Regression test	Ensures a new change did not break previously working functionality.

---

47. Automated testing

One strength of the project is extensive automated testing.

The final MACS/operator-state work reached:

186 tests passed

Earlier project phases also had progressively increasing test suites.

Explain:

“Every major subsystem was validated independently first, then integrated, and regression tests were retained as the project grew.”

---

48. Test types

Keyword	Meaning
Unit test	Tests one function/component.
Integration test	Checks communication between multiple components.
Regression test	Ensures old functionality still works after a change.
Fixture	Controlled test inputs used to reproduce states.
Live Gazebo validation	Tests actual integrated ROS + simulator behavior.

Do not claim a fixture is the same as a full physical simulation campaign.

---

49. Demo runner

Your main scenario commands represent different system configurations:

./scripts/run_marine_demo.sh basic --gui

./scripts/run_marine_demo.sh pid --gui --rviz

./scripts/run_marine_demo.sh mission --gui --rviz

./scripts/run_marine_demo.sh cooperative --gui --rviz

Explain:

basic

→ vehicle/environment

pid

→ path following

mission

→ perception + autonomous target mission

cooperative

→ leader/follower formation

---

50. What happens in your PID demo

Explain it in this sequence:

1. Gazebo starts.

2. USV model appears.

3. Reference path is generated.

4. Odometry gives actual vehicle pose.

5. LOS computes desired heading.

6. PID compares desired and actual heading.

7. Mixer creates port/starboard thrust.

8. Vessel moves.

9. Feedback repeats continuously.

10. MACS displays CTE, heading, path progress and thrust.

That is an excellent explanation to give while showing the Navigation screen.

---

51. What happens in the mission demo

1. USV begins search.

2. Camera publishes images.

3. YOLO processes images.

4. Target detection is temporally confirmed.

5. Mission FSM changes from search to acquisition.

6. Bearing guides target-centering.

7. Depth-derived range determines distance.

8. Vehicle approaches target.

9. Target loss triggers reacquisition if needed.

10. 12 m stand-off is reached.

11. Propulsion is set to zero.

12. Inspection/final image is recorded.

13. Mission enters terminal state.

---

52. What happens in cooperative demo

1. Leader and follower USVs start.

2. Each has independent odometry/controller state.

3. Leader follows its reference path.

4. Leader state is shared.

5. Coordinator determines follower target position.

6. Desired in-trail spacing = 8 m.

7. Follower controller reduces formation error.

8. MACS shows actual spacing and spacing error.

---

53. Why ROS 2 was important

A useful viva answer:

“ROS 2 allowed us to separate navigation, control, perception, mission logic, cooperative coordination and GUI into independent nodes. Because they communicate through standard interfaces, individual modules can be tested or replaced without rewriting the entire system.”

---

54. Why simulation first?

“Marine field trials are expensive, risky and difficult to repeat. Gazebo allows deterministic, repeatable validation of navigation, perception, controller and mission logic before moving toward physical hardware.”

---

55. Why a hierarchical architecture?

Your architecture is effectively:

Mission Layer

↓

Guidance Layer

↓

Control Layer

↓

Actuation Layer

with perception feeding the mission layer.

Meaning:

Mission decides WHAT to do.

Guidance decides WHERE to point.

Controller decides HOW to correct error.

Thrusters physically execute it.

This is a very strong way to explain the project.

---

56. Why PID and SMC?

“PID provides a simple, interpretable baseline controller. SMC provides a nonlinear robust-control alternative. Implementing both allowed us to evaluate different control approaches using the same vehicle, paths and metrics.”

---

57. Why YOLO?

“YOLO was selected because it performs object detection in a single inference pass and is suitable for real-time perception. The nano P6 configuration was chosen because the project ran primarily on constrained inference resources during validation.”

---

58. Why separate range from YOLO?

This is a particularly good design answer:

“Bounding-box size is not a reliable physical distance measurement. Therefore object classification/localization and range estimation were separated. YOLO identifies where the vessel is in the image, while the independent depth channel provides range.”

---

59. Why temporal confirmation?

“A single detection can be noise or a false positive. Temporal confirmation requires consistent evidence before the mission changes behavior.”

---

60. Why reacquisition?

“A marine target can temporarily disappear due to detector variation, orientation or viewing conditions. Reacquisition prevents the mission from immediately failing after a temporary loss.”

---

61. Why Safe Stop?

“Autonomous systems require an independent supervisory safety path. Safe Stop allows the operator to disable propulsion regardless of the current mission state and then verifies that thrust has actually reached zero.”

---

62. Current limitations

Do not hide these if your guide asks.

Your main limitations are:

- simulation rather than real sea trials;
- visual water waves do not physically affect dynamics;
- direct-force thruster model rather than detailed propeller hydrodynamics;
- pseudo-thermal rather than calibrated LWIR;
- detector class mapping remains generic VESSEL;
- occasional perception loss can occur;

- current cooperative system uses two vessels;
- completed navigation requires fresh simulation for another path;
- no physical gimbal integrated yet;
- real sensors/hardware remain future work.

---

63. Future work

You can say:

“Future work would integrate a real USV, GPS/IMU, calibrated camera or thermal camera, gimbal control, real thruster drivers, more realistic nonlinear wave/current models, larger multi-agent formations, and hardware-in-the-loop/sea trials.”

---

64. Your complete project story in simple language

If your guide says **“Explain your whole project”**, say approximately:

“I started by creating a marine environment in Gazebo Harmonic and validating buoyancy and the catamaran USV. Then I added differential thrust, odometry, IMU and TF. I generated smooth straight, curved and S-shaped reference paths using cubic splines and used LOS guidance to calculate the desired heading. For path tracking, I implemented PID and SMC controllers and evaluated them using cross-track and heading-error metrics.

After navigation was stable, I integrated a camera and YOLO-based vessel detector. RGB detection was unreliable with the available trained model, so the final simulation used an assigned-temperature thermal-style perception pipeline. Detection is temporally confirmed, and target bearing is obtained from its image position. Distance is obtained independently from depth information.

I then built an autonomous mission FSM that performs search, target acquisition, approach, target-loss recovery, reacquisition and stand-off arrival. At approximately 12 metres, propulsion is brought to zero before inspection.

The framework was then extended to two USVs for cooperative leader-follower control with an 8-metre in-trail spacing target.

Finally, I developed MACS, which provides navigation, perception, cooperative-control, research and safety information in a single operator GUI. It has health monitoring, lifecycle-aware state handling, mission-start protection and a verified Safe Stop function.

The entire framework is modular through ROS 2 and has been validated with automated tests and live Gazebo scenarios.”

---

65. The five most important numbers to remember

PID gains

Kp = 0.5

Ki = 0

Kd = 0.6

Mission stand-off

12.0 m

Successful final arrival

≈ 11.79 m

Cooperative spacing

8.0 m

Latest MACS tests

186 passed

Also remember:

YOLO model:

YOLOv5n6 / P6

~3.11 M parameters

640 input

10 classes

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66. Five sentences that prevent common viva mistakes

Say these exactly if necessary:

“Gazebo performs the simulation; RViz visualizes ROS data.”

“ROS 2 is the middleware connecting the modules.”

“YOLO detects the vessel, but YOLO does not provide our range measurement.”

“The animated ocean surface is visual; physical wave dynamics were not modeled by that visual mesh.”

“Safe Stop means verified zero propulsion; it does not mean the boat instantaneously reaches zero velocity.”

And for PID versus SMC:

“Both were evaluated; I do not claim that SMC globally outperforms PID.”

Final mental map

Remember just this chain and you can reconstruct almost the entire project during your explanation:

GAZEBO MARINE WORLD

↓

CATAMARAN USV

↓

ODOMETRY + IMU + TF

↓

CUBIC-SPLINE PATH

↓

LOS GUIDANCE

↓

PID / SMC

↓

DIFFERENTIAL THRUST

↓

PATH FOLLOWING

↓

CAMERA

↓

YOLO VESSEL DETECTION

↓

TEMPORAL CONFIRMATION

↓